Queueing analysis of a battery-backed electrical vehicle charging system

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ABSTRACT

We study a battery-buffered electric vehicle fast charging station modeled as a piecewise-deterministic Markov process. A battery mediates between the grid connection and one or more chargers, enabling fast service when the battery is charged and falling back to the grid rate when depleted. For the single-server loss system we derive the stationary distribution in closed form, uncovering a decoupling property whereby the charger state and battery level are independent in steady state, and obtain explicit formulas for several performance metrics. We extend the analysis to the multi-server loss system, showing that the stationary densities retain an exponential form with occupancy weights governed by a simple recurrence, and that the classical Erlang-B formula is recovered at the critical load where the mean battery drift vanishes.

Keywords

Piecewise deterministic Markov processes, Electrical Vehicles.

1. INTRODUCTION

The rapid adoption of electric vehicles (EVs) is placing increasing demands on electrical infrastructure. A key challenge is the provisioning of public *fast charging*: modern DC fast chargers can deliver 150 kW or more per port, requiring grid connections sized for their peak output. Yet in practice a charging station is not used continuously; vehicles arrive stochastically, and the chargers are idle most of the time. Sizing the grid connection for worst-case simultaneous demand is therefore wasteful, civil infrastructure becomes costly, and utility demand charges (fees based on peak power draw) can represent the dominant operating cost of a fast charging site [8]. A battery, acting as an energy buffer, placed between the grid and the chargers resolves this mismatch: the battery gradually charges from the grid at a steady, lower rate and discharges at high power during the brief intervals a vehicle is present, enabling a much smaller and cheaper grid connection. While battery storage at EV stations has been studied before, it has been done primarily in the context of renewable energy integration, where the battery smooths intermittent solar or wind supply [1,6,10]. However, with the advent of fast DC chargers available in highways and remote areas, using a battery buffer in these low-utilization regimes has become equally compelling for purely grid-connected stations.

The central analytical question this raises is a queueing one: given that vehicles arrive randomly and the battery recharges during idle or low utilization periods, how often does a vehicle arrive to find the battery depleted, forcing service at the slower grid rate? This coupling between the continuous battery state and the discrete service rate is what distinguishes the battery-buffered station from a standard queue, and it has not been captured in the existing EV charging literature, which models charging with a fixed service rate [2, 5, 9].

In this paper, we propose a station model using a Piecewise-Deterministic Markov Process (PDMP) [3, 4]: the battery level evolves deterministically between random events, while car arrivals and departures trigger instantaneous switches. The state is then the pair (n, x) of system occupancy and battery level. We derive the generator of this process, and compute the stationary distribution in closed form for the single-server loss system. A key structural finding is a *decoupling property*: in steady state the charger occupancy and battery level are conditionally independent given the battery is the interior of the state space (partially charged), so that the stationary density factors as a product of an exponential in x and a mode-dependent weight. This yields explicit formulas for the blocking probability, mean charging time, and a *slowdown factor* quantifying the service degradation as a function of a car's energy demand. We then extend the analysis to a loss system with s parallel chargers sharing a common battery. The exponential form of the stationary density is preserved, with the mode weights now governed by a simple recurrence. Armed with these formulas, we discuss several performance issues and design objectives that provide guidelines on how to design such systems.

The paper is organized as follows. Section 2 introduces the single-server model and its PDMP formulation. Section 3 derives the stationary distribution and performance metrics. Section 4 treats the many-server case. Section 5 concludes.

2. SINGLE-SERVER MODEL

We consider a single EV charging station consisting of one charger connected to the electrical grid, equipped with a battery buffer of capacity $B > 0$. Cars arrive according to a Poisson process of rate $\lambda > 0$, each carrying an energy

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demand $Y \sim \mathrm{Exp}(\mu)$. The station operates as a *loss system*: a car that arrives when the charger is busy is turned away.

The system parameters are the following: the grid may supply a maximum power p_1, and this energy may be used to charge the battery or the car, if present. When the system is idle, the battery is recharged at rate $0 < p_0 \leqslant p_1$. Inequality reflects that we may want to recharge the battery at a slower rate to preserve battery life. The EV charging station itself may provide power $p_2 > p_1$, which amounts to fast charging. When a car is present, this power is provided by the grid at rate p_1 and the extra amount $c := p_2 - p_1 > 0$ is provided by the battery, whenever is not empty. If the battery empties, the system reverts to charging only from the grid at rate p_1, and the battery stays depleted until the current car finishes service. We denote by $x(t) \in [0, B]$ the current state of the battery at time t, and by $n(t) \in \{0, 1\}$ the presence of not of a charging vehicle.

Under the above assumptions, when a car is admitted, service proceeds in at most two phases. In *Phase 1* (fast charging), while the battery level $x > 0$, the car charges at rate p_2 and the battery discharges at rate c. In *Phase 2* (slow charging), once the battery is exhausted, $x = 0$ and the car charges at rate p_1, while x remains at zero. A car with small demand may complete service entirely in Phase 1 if its service requirement is low enough. Under the exponential energy demands assumed before, the instantaneous departure rate from the system is therefore $\mu p(x)$, where:

$$p(x) = p_2 \, \mathbf{1}_{x>0} + p_1 \, \mathbf{1}_{x=0}. \tag{1}$$

The system state at time t is thus the pair $Z(t) = (n(t), x(t))$. Between arrival and departure events, the battery evolves deterministically according to $\dot{x} = r(n, x)$, where

$$r(n,x) = \begin{cases} p_0 & n = 0,\ x < B, \\ 0 & n = 0,\ x = B, \\ -c & n = 1,\ x > 0, \\ 0 & n = 1,\ x = 0. \end{cases} \tag{2}$$

The jumps are spontaneous events: *arrivals* at $(0, x)$ occur at rate λ and the states transitions to $(1, x)$; *departures* at $(1, x)$ with $x > 0$ occur at rate μp_2 and transition to $(0, x)$; *departures* at $(1, 0)$ occur at rate μp_1 and transition to $(0, 0)$, or in short $(1, x) \mapsto (0, x)$ at rate $\mu p(x)$. In all cases the battery level is unchanged at the jump epoch.

The process $Z(t)$ with state space $\{0, 1\} \times [0, B]$ fits the PDMP framework of Davis [3,4], though the standard theory requires an open state space and forces the process to jump before reaching the boundary. Neither condition holds here: E is compact, and the flow reaches $x = 0$ and $x = B$ in finite time without a jump. Nevertheless, the vector field (2) is bounded and monotone on each mode, so the flow is globally well-defined. The strong Markov property follows from the martingale-problem approach, which extends to this setting since all jump rates are bounded and the jump kernel leaves x unchanged, i.e. jumps in the continuous variable are bounded.

Note that an arriving car always finds $x > 0$ with probability one. During any idle period the battery charges at rate $p_0 > 0$, and since the idle time before the next arrival is $\mathrm{Exp}(\lambda)$ (positive a.s.), the battery is strictly positive at every arrival epoch. The state $(n = 0, x = 0)$ therefore should have zero probability under any stationary distribution.

The extended generator $\mathcal{A}$ of $Z(t)$ has domain $\mathcal{D}(\mathcal{A})$ consisting of functions $\psi : E \to \mathbb{R}$ such that $\psi(n, \cdot)$ is absolutely continuous on $[0, B]$ for each n. For $\psi \in \mathcal{D}(\mathcal{A})$,

$$\mathcal{A}\psi(0,x) = r(0,x)\,\psi_0'(x) + \lambda\,[\psi(1,x) - \psi(0,x)], \tag{3}$$

$$\mathcal{A}\psi(1,x) = r(1,x)\,\psi_1'(x) + \mu p(x)\,[\psi(0,x) - \psi(1,x)], \tag{4}$$

where $\psi_n' = \partial_x \psi(n, \cdot)$ is defined Lebesgue-a.e. The first term in each line is the flow contribution; the second is the jump contribution. For any $\psi \in \mathcal{D}(\mathcal{A})$, the process $M_t^\psi := \psi(Z(t)) - \psi(Z(0)) - \int_0^t \mathcal{A}\psi(Z(s))\,ds$ is a local martingale [3].

The adjoint operator $\mathcal{A}^*$ operates on the probability distributions on E and provides the forward Kolmogorov equation of the system [4]. A stationary distribution π satisfies $\int \mathcal{A}\psi \, d\pi = 0$ for all $\psi \in \mathcal{D}(\mathcal{A})$, which is the weak form of the Kolmogorov forward equation $\mathcal{A}^*\pi = 0$. We now proceed to solve the steady state of this system by positing a particular form of π using the characteristics of the system.

3. SINGLE-SERVER IN STEADY-STATE

Since E is compact and $Z(t)$ is irreducible (every open rectangle $\{n\} \times (a, b)$ is reachable with positive probability from any starting point, by a combination of deterministic drift and Poisson events), the process is ergodic with a unique stationary distribution π [7].

We seek π as a mixed measure with densities on $(0, B)$ and point masses at the boundary points where flow accumulates. Substituting this form into $\int \mathcal{A}\psi \, d\pi = 0$ and integrating by parts yields a system of ODEs on $(0, B)$ and flux-balance conditions at the boundary. In particular, we look for solutions π in the mixed form

$$\pi = f_0(x)\,dx + f_1(x)\,dx + \pi_{0,B}\,\delta_{(0,B)} + \pi_{1,0}\,\delta_{(1,0)}, \tag{5}$$

where $f_n \geq 0$ are densities on $(0, B)$ and $\pi_{0,B}, \pi_{1,0} \geq 0$ are point masses at the corners where the flow accumulates: $x = B$ in mode 0 (since $r(0, \cdot) = p_0 > 0$) and $x = 0$ in mode 1 (since $r(1, \cdot) = -c < 0$). The other two corners should carry no atoms.

Steady-state equations. Setting $\mathcal{A}^*\pi = 0$ and using (3)–(4), the stationarity condition $\int \mathcal{A}\psi \, d\pi = 0$ for all $\psi \in \mathcal{D}(\mathcal{A})$ yields, after integration by parts, the following system. The *interior equations* on $(0, B)$ are

$$p_0\,f_0'(x) = -\lambda\,f_0(x) + \mu p_2\,f_1(x), \tag{6}$$

$$-c\,f_1'(x) = \quad \lambda\,f_0(x) - \mu p_2\,f_1(x). \tag{7}$$

The *boundary flux equations* are obtained by flux conservation at each boundary point: mass arriving at a boundary via the flow must equal mass leaving via jumps. At $x = 0$ in mode 1, the inward flow $c\,f_1(0^+)$ feeds the atom $\pi_{1,0}$, which drains via departures at rate μp_1:

$$c\,f_1(0^+) = \mu p_1\,\pi_{1,0}. \tag{8}$$

At $x = 0$ in mode 0, the atom $\pi_{1,0}$ is created by departures and immediately carried into the interior by the flow at rate p_0:

$$p_0\,f_0(0^+) = \mu p_1\,\pi_{1,0}. \tag{9}$$

At $x = B$ in mode 0, the inward flow $p_0\,f_0(B^-)$ feeds the atom $\pi_{0,B}$, which drains via arrivals at rate λ:

$$p_0\,f_0(B^-) = \lambda\,\pi_{0,B}. \tag{10}$$

At $x = B$ in mode 1, arrivals from $\pi_{0,B}$ feed the interior flow at rate λ:

$$c\,f_1(B^-) = \lambda\,\pi_{0,B}. \tag{11}$$

These are supplemented by normalization: $\int_0^B (f_0 + f_1)\,dx + \pi_{0,B} + \pi_{1,0} = 1$.

We can now state the main result of this section.

THEOREM 3.1. *Let $\theta = \mu p_2/c - \lambda/p_0$. The unique stationary distribution of $Z(t)$ has the form* (5) *with*

$$f_0(x) = \frac{e^{\theta x}}{K}, \qquad f_1(x) = \frac{p_0}{c} \cdot \frac{e^{\theta x}}{K}, \qquad x \in (0, B), \quad (12)$$

$$\pi_{1,0} = \frac{p_0}{\mu p_1\, K}, \qquad \pi_{0,B} = \frac{p_0\, e^{\theta B}}{\lambda\, K}, \quad (13)$$

where the normalization constant is

$$K = \frac{p_0 + c}{c} \cdot \frac{e^{\theta B} - 1}{\theta} + \frac{p_0}{\mu p_1} + \frac{p_0\, e^{\theta B}}{\lambda}, \quad (14)$$

with the convention $(e^{\theta B} - 1)/\theta = B$ when $\theta = 0$.

PROOF SKETCH. **Decoupling.** Adding (6) and (7) gives $\frac{d}{dx}[p_0 f_0 - c f_1] = 0$, so $p_0 f_0 - c f_1$ is constant on $(0, B)$. Equations (8) and (9) together give $p_0 f_0(0^+) = c f_1(0^+)$, so the constant is zero: $f_1(x) = (p_0/c)\, f_0(x)$ for all $x \in (0, B)$.

Exponential solution. Substituting $f_1 = (p_0/c) f_0$ into (6) yields $f_0'(x) = \theta\, f_0(x)$ with $\theta = \mu p_2/c - \lambda/p_0$, whose solution is $f_0(x) = f_0(0^+)\, e^{\theta x}$.

Atoms and normalization. The atoms follow directly from (9) and (10). One can verify that (11) is automatically satisfied, confirming consistency. Setting $f_0(0^+) = 1/K$ and imposing normalization determines K as in (14).

Boundary conditions at $x = B$. Equations (10)–(11) are both satisfied by the exponential solution with the atoms given in (13), confirming that no additional constraint on θ arises: the parameter θ is determined entirely by the interior ODE, not by the boundary. □

The parameter $\theta = \mu p_2/c - \lambda/p_0$ has a precise physical interpretation. Consider a single idle-busy cycle in the absence of battery boundaries: the battery charges at rate p_0 during the idle period (mean duration $1/\lambda$) and discharges at rate c during the busy period (mean duration $1/(\mu p_2)$). The expected net battery change per cycle is

$$\Delta = \frac{p_0}{\lambda} - \frac{c}{\mu p_2},$$

One verifies directly that $\theta > 0$ if and only if $\Delta > 0$: both are equivalent to $\lambda/\mu < p_0 p_2/c$. Thus θ is positive when the offered load λ/μ is below the threshold $p_0 p_2/c$, meaning the battery gains energy on average per cycle, and negative when it loses energy. Thus $\theta > 0$ corresponds to a lightly loaded station where the battery tends to fill, mass concentrates near $x = B$, and blocking is close to the M/M/1/1 value at the fast rate p_2. When $\theta < 0$ the battery tends to drain on average, mass concentrates near $x = 0$, and a significant fraction of service occurs at the slow rate p_1. When $\theta = 0$ the expected gain per cycle is exactly zero and the stationary density is uniform on $(0, B)$.

We now derive the key performance metrics directly from Theorem 3.1.

PROPOSITION 3.2. *The stationary blocking probability of the system is*

$$P_B = \int_0^B f_1(x)\,dx + \pi_{1,0} = \frac{p_0}{K}\left(\frac{e^{\theta B} - 1}{c\,\theta} + \frac{1}{\mu p_1}\right). \quad (15)$$

This follows directly by the PASTA property of arrivals. Note that this probability is also the probability that the charger is working. As $B \to 0$, $P_B \to \lambda/(\lambda + \mu p_1)$, the blocking probability of an $M/M/1/1$ queue at rate p_1. As $B \to \infty$ with $\theta > 0$, $P_B \to \lambda/(\lambda + \mu p_2)$, corresponding to service always at the fast rate p_2. When $\theta < 0$, P_B converges to a value strictly between these two limits even as $B \to \infty$, reflecting the residual probability of finding the battery depleted.

PROPOSITION 3.3. *The mean service time of an admitted car is:*

$$\mathbb{E}[T] = \frac{P_B}{\lambda(1 - P_B)}. \quad (16)$$

This follows directly from Little's law and the fact that only one car is present with probability P_B.

PROPOSITION 3.4 (SLOWDOWN FACTOR). *Given admission at level x with demand $Y = y$, let $T(x, y) = y/p_2 + (p_1^{-1} - p_2^{-1}) \max(y - p_2 x/c, 0)$ be the charging time. The* slowdown factor *$S(y) = \mathbb{E}[T \mid Y = y]/(y/p_2) \geq 1$ quantifies degradation relative to ideal fast charging. With $\tilde{x}(y) = cy/p_2$ the critical battery level:*
For $y < p_2 B/c$ (battery may survive):

$$S(y) = 1 + \frac{p_2 - p_1}{p_1\, y\, K(1 - P_B)\,\theta}\left[\frac{p_2}{c\,\theta}\left(e^{\theta c y/p_2} - 1\right) - y\right]. \quad (17)$$

For $y \geq p_2 B/c$ (battery always depletes):

$$S(y) = 1 + \frac{p_2 - p_1}{p_1\, y\, K(1 - P_B)}\left\{\frac{1}{\theta}\left[(y - \tfrac{p_2 B}{c})e^{\theta B} - y + \tfrac{p_2(e^{\theta B} - 1)}{c\theta}\right] + \tfrac{p_0 e^{\theta B}}{\lambda}(y - \tfrac{p_2 B}{c})\right\}. \quad (18)$$

$S(y)$ satisfies $S(y) \geq 1$, is increasing in y, tends to 1 as $y \to 0$ or $B \to \infty$, and tends to p_2/p_1 as $y \to \infty$ or $B \to 0$.

4. THE MANY-SERVER CASE

We now extend the model to $s \geq 1$ parallel chargers sharing a single battery buffer of capacity B and a single grid connection of rate p_1. The station continues to operate as a loss system: a car that arrives when all s chargers are busy is turned away.

Assumptions. We impose $p_1 < s p_2$, so the grid alone cannot sustain all s chargers simultaneously at full rate; the battery is essential at full occupancy. We also set $p_0 = p_1$, so the idle recharge rate equals the grid supply rate; this simplifies the structure of the vector field.

State space and dynamics. The state is $Z(t) = (n(t), x(t))$ with $n(t) \in \{0, \ldots, s\}$ busy chargers and $x(t) \in [0, B]$; $E = \{0, \ldots, s\} \times [0, B]$ is compact. With n busy chargers each drawing p_2 and the grid supplying p_1, the net drift is $r_n = p_1 - n p_2$. Setting $p_0 = p_1$ makes this hold for all n: the battery charges at rate p_1 when idle and drains at rate $n p_2 - p_1$ when $n > p_1/p_2$. The vector field is

$$\dot{x} = \begin{cases} r_n & 0 < x < B,\ r_n \neq 0, \\ 0 & \text{otherwise (boundary reflection).} \end{cases} \quad (19)$$

Define the *critical occupancy* $n^* = \lfloor p_1/p_2 \rfloor \in \{0, \ldots, s - 1\}$; the battery charges when $n \leq n^*$ and drains when $n > n^*$.

The jumps are: arrivals at rate λ when $n < s$, transitioning $n \to n + 1$ with x unchanged; departures at total rate $n\mu p_2$

when $x > 0$, and at rate μp_1 when $x = 0$ (the grid supplies p_1 shared equally, giving total throughput μp_1 regardless of occupancy), each transitioning $n \to n-1$ with x unchanged.

The process $Z(t)$ is again a PDMP on a compact state space, and the same martingale-problem argument as in Section 2 establishes the strong Markov property. Irreducibility follows by the same reachability argument, so $Z(t)$ is ergodic with a unique stationary distribution.

Form of the stationary distribution. The flow direction at each boundary point now depends on the occupancy level n. Atoms accumulate at (n, B) for $n \le n^*$ (where the flow pushes toward $x = B$) and at $(n, 0)$ for $n > n^*$ (where the flow pushes toward $x = 0$). We therefore seek a stationary distribution of the form

$$\pi = \sum_{n=0}^{s} f_n(x)\,dx + \sum_{n=0}^{n^*} \pi_n^{(B)}\,\delta_{(n,B)} + \sum_{n=n^*+1}^{s} \pi_n^{(0)}\,\delta_{(n,0)}, \quad (20)$$

where $f_n \ge 0$ are densities on $(0, B)$ and $\pi_n^{(B)}, \pi_n^{(0)} \ge 0$ are point masses.

Interior equations. Setting $\mathcal{A}^*\pi = 0$ on $(0, B)$ gives, for $0 \le n \le s$,

$$r_n f_n'(x) = \lambda f_{n-1}(x)\,\mathbf{1}_{n>0} + (n+1)\mu p_2\, f_{n+1}(x)\,\mathbf{1}_{n<s} - (\lambda\,\mathbf{1}_{n<s} + n\mu p_2)\, f_n(x). \quad (21)$$

THEOREM 4.1. *Let $\rho = \lambda/(\mu p_2)$. There exists a unique $\theta \in \mathbb{R}$, determined by the boundary flux equations together with normalization, such that the stationary densities take the form*

$$f_n(x) = C\,a_n(\theta)\,e^{\theta x}, \qquad x \in (0, B),\ n = 0, \dots, s, \quad (22)$$

where $C > 0$ is a normalization constant and the coefficients $a_n(\theta)$ satisfy the recurrence $a_0 = 1$ and

$$a_{n+1} = \frac{\theta\, S_n + \lambda\, a_n}{(n+1)\mu p_2}, \qquad S_n = \sum_{k=0}^{n} r_k\, a_k, \quad n = 0, \dots, s-1. \quad (23)$$

When $\theta = 0$, the recurrence yields $a_n = \rho^n/n!$ and the distribution reduces to the classical Erlang-B form with uniform density in x. For $\theta \ne 0$, the blocking probability

$$P_B = C\,a_s\,\frac{e^{\theta B} - 1}{\theta} + \pi_s^{(0)}\,\mathbf{1}_{n^*<s} \quad (24)$$

deviates from the Erlang-B value, with the battery shifting mass toward $x = B$ when $\theta > 0$ and toward $x = 0$ when $\theta < 0$.

PROOF SKETCH. Summing (21) from $k = 0$ to n telescopes the right side to $-\lambda f_n + (n+1)\mu p_2 f_{n+1}$. The exponential form $f_n = C a_n e^{\theta x}$ then yields (23). The value of θ is fixed by the boundary flux equations together with normalization. For $s = 1$ and $p_0 = p_1$, one recovers $\theta = \mu p_2/c - \lambda/p_1$ of Theorem 3.1. □

The parameter θ admits the same physical interpretation as in the single-server case. In the unconstrained system (infinite battery), the occupancy evolves as an $M/M/s/s$ queue with Erlang-B stationary weights $\pi_n \propto \rho^n/n!$, giving a mean battery drift $\bar{r} = p_1 - (\lambda/\mu)(1 - P_B^{\text{Erlang}})$. The sign of $\bar{r}$ determines the sign of θ: positive when the grid can sustain the average load (battery fills in the long run) and negative otherwise. The case $\theta = 0$ corresponds exactly to $\bar{r} = 0$, at which point the Erlang-B distribution is recovered.

Performance metrics. The blocking probability is given by (24). By PASTA and Little's law the mean charging time follows from P_B exactly as in Propositions 3.2 and 3.3. The slowdown factor $S(y)$ generalizes directly: the admitted battery level has the same exponential density $f_0(x)/(1 - P_B)$ on $(0, B)$ (now with s-server normalization), and the conditional charging time $T(x, y)$ is unchanged, so the formulas of Proposition 3.4 apply with the updated P_B and $K = 1/C$.

5. CONCLUSIONS

We have studied a battery-buffered EV fast charging station using the framework of piecewise-deterministic Markov processes. For the single-server loss system we derived the stationary distribution in closed form, establishing a decoupling property that separates the battery level from the charger state in steady state, and obtained explicit expressions for the blocking probability, mean charging time, and slowdown factor. We extended the analysis to a loss system with s parallel chargers sharing a common battery, showing that the stationary densities retain an exponential form in the battery level with occupancy-dependent weights governed by a simple recurrence. In both cases the key parameter θ, the sign of the mean battery drift, determines whether the system operates in a battery-rich or battery-scarce regime and controls the deviation of the blocking probability from the classical Erlang-B value.

The analysis highlights a structural difference from standard fluid queues: the state-dependent service rate at $x = 0$ couples the discrete and continuous components in a way that prevents a product-form solution in the multi-server case while still admitting an exponential form in x. Numerical illustrations of the performance metrics and a study of the optimal battery sizing problem are left for future work.

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